

JIPING WANG

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SUMMARY

Ph.D. Candidate in Statistics with industry experience in data science through internships at Genentech and Ocular Therapeutix. Experienced in statistical modeling, machine learning, Python, and SQL, with applications in clinical trial analytics and experimentation. Passionate about developing rigorous, practical solutions for real-world data science problems.

EDUCATION

Florida State University

Ph.D. Candidate in Statistics (*GPA: 3.98/4.00*)

Tallahassee, FL.

Aug. 2020 - Present

Xiamen University

M.S. in Mathematical Statistics (*GPA: 3.71/4.00; Rank: 1/17*)

Xiamen, China

Sep. 2017 - Jun. 2020

B.S. in Mathematical Statistics (*GPA: 3.77/4.00; Rank: 2/31*)

Sep. 2013 - Jun. 2017

PROFESSIONAL EXPERIENCE

Product Development and Data Science Intern

May 2025 – Nov 2025

Genentech

- Simulated visual field data to expand limited PRODIGIES glaucoma datasets with small sample sizes and short duration, enabling large-scale trial analysis and robust endpoint evaluation.
- Quantified trade-offs between trial designs and demonstrated that cluster-based designs reduce required sample size by 40%, directly influencing cost and timeline decisions.
- Developed slope-based disease progression endpoints outperforming FDA-approved event-based metrics, enabling earlier detection and faster decision-making.
- Implemented longitudinal modeling workflows (PLR, Hierarchical Bayesian Models, AGIS) to analyze MD and VFI trajectories for glaucoma progression.

Biostatistics Intern

May 2024 – Aug 2024

Ocular Therapeutix

- Simulated efficacy outcomes (BCVA, CSFT) across diverse scenarios to support protocol amendments and endpoint selection.
- Applied advanced missing-data strategies (Pattern Mixture Models, LOCF) to improve robustness of primary analyses.
- Conducted causal inference analyses using linear mixed models, delivering insights that informed clinical and regulatory decisions.

SELECTED PROJECTS

Hierarchical Sure Independence Screening for Multiresolution Classification Aug. 2023 – Present

- Invented HierSIS, which lifts sure independence screening from flat to hierarchically structured multinomial responses by screening separately at each node of the class hierarchy, with sure screening guarantees generalizing Fan & Song (2010).
- Demonstrated higher true-signal recovery than flat SIS and penalized regression competitors in simulation studies across $p \in \{1500, 2000, 3000, 5000\}$ predictors.
- Applied to the Hao et al. (2021) PBMC scRNA-seq dataset (152,244 cells, 27,373 genes, 24 fine cell types nested in 11 coarse classes), improving macro-F1 by 0.02 and balanced error rate by 0.03 over the leading competitor (mrMLR).

Matrix-Response Regression with Low Tubal Rank and Sparsity (*Manuscript*) Jun. 2022 – Dec 2025

- Proposed a matrix-response regression framework integrating low tubal-rank and sparsity constraints for structured high-dimensional data.

- Designed proximal gradient and linearized ADMM algorithms with convergence guarantees; proved non-asymptotic error bounds and variable selection consistency under restricted strong convexity.
- Benchmarked against 6 competing methods across 9 simulation settings, consistently achieving lower estimation error, better rank recovery, and variable selection consistency across sparsity structures.
- Applied to EEG seizure data (19 electrodes, 60 timepoints), pinpointing electrodes P4, C4, and T4 as the strongest seizure biomarkers, consistent with established neurological findings.

Censored Mean Variance Sure Independence Screening for Ultra-high Dimensional Survival Data
 Mar. 2018 – Oct. 2019

- Developed a model-free feature screening method for ultra-high-dimensional survival data under censoring, with rigorous theoretical guarantees
- Established sure screening and ranking consistency properties under mild regularity conditions, supported by formal proofs.
- Conducted extensive simulation studies demonstrating near-100% true-variable inclusion rates under 20%–40% censoring.
- Applied the method to breast cancer survival data, achieving the strongest overlap with consensus gene sets and the lowest Kolmogorov Smirnov distance and Cramer-von Mises distance for Cox model compared with competing methods.

TEACHING EXPERIENCE

Instructor of Record, Florida State University Spring 2022; Fall 2023; Spring 2024

- Taught STA 1013 (Statistics Through Examples) as sole instructor for 3 semesters; responsible for course design, lectures, and grading.
- Used real-world datasets and R to connect statistical theory to applied reasoning and interpretation.

Teaching Assistant, Florida State University Aug. 2021 – Present

- Assisted in undergraduate and graduate courses covering regression, probability, and statistical computing.
- Held office hours and graded assignments for courses using R, Python, and MATLAB.

PUBLICATIONS

Jiping Wang, Xin Zhang. “Matrix-Response Regression with Low Tubal Rank and Sparsity”. *In Submission*.
 Wei Zhong, **Jiping Wang**, Xiaolin Chen. “Censored mean variance sure independence screening for ultrahigh-dimensional survival data”. *Computational Statistics & Data Analysis* 2021.

TECHNICAL SKILLS

Programming: Python, SQL, R, MATLAB
Machine Learning: Statistical Learning, Deep Learning, Large Language Models (LLMs), Natural Language Processing (NLP)
Statistics: A/B Testing, Causal Inference, Bayesian Modeling, Missing Data, High-Dimensional Statistics
Optimization: ADMM, Proximal Gradient Methods, Large-Scale Simulation
Frameworks: PyTorch, scikit-learn, NumPy, Pandas, Transformers, TRL, PEFT, AWS, Git

HONORS

American Statistical Association Florida Chapter Student Presentation Award	Mar. 2024
Clara Kibler Davis Scholarship	Oct. 2022
Graduation with Honors (Cum Laude)	Jun. 2020
Graduation with Honors (Cum Laude)	Jun. 2017